MATH40004 - Calculus and Applications

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More details at https://stones-y.github.io.

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Introduction

This module assumes mastery of A-Level Mathematics (and Further Mathematics), in particular differentiation, integration and associated techniques, curve sketching, and elementary mechanics. Ultimately, this is a methods-based course, and while a certain level of mathematical communication and clarity is expected, we do not focus on rigorously proving facts about calculus in this course (that is the purpose of MATH40002, Analysis I). Any proofs that are omitted will be covered in MATH40002 (most of the time, in this course, we are just interested in the result and the consequences).

Chapter 1

Limits and Continuity

1.1 Limits of Functions

The fundamental concept of calculus is the limit. We formalise the notion of a function approaching a specific value. All of this content is covered in depth in Analysis I (MATH40002) so we will not spend much time on it, and all proofs can be found in Analysis I.

Definition 1.1 (ε-δ definition of a limit)**.** Let f be a function defined on an open interval containing x_0, except possibly at x_0 itself. We say that $\lim_{x\to x_0} f(x) = \ell$ if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that for all x,

$$0 < |x - x_0| < \delta \implies |f(x) - \ell| < \varepsilon.$$

Example 1.2. We prove that $\lim_{x\to 2}(3x - 1) = 5$. Given $\varepsilon > 0$, we want to find $\delta > 0$ such that $0 < |x-2| < \delta$ implies $|(3x-1)-5| < \varepsilon$. We manipulate the absolute value we are trying to bound:

$$|(3x - 1) - 5| = |3x - 6| = 3|x - 2|.$$

To ensure this is strictly less than ε, we require $3|x - 2| < \varepsilon$, which is equivalent to $|x - 2| < \frac{\varepsilon}{3}$. Therefore, we choose $\delta = \frac{\varepsilon}{3}$. Then for all x satisfying $0 < |x - 2| < \delta$, we have $|(3x - 1) - 5| = 3|x - 2| < 3\delta = \varepsilon$.

Theorem 1.3 (Algebra of Limits)**.** *Suppose* $\lim_{x\to x_0} f(x) = L$ *and* $\lim_{x\to x_0} g(x) = M$*. Then:*

1. $\lim_{x\to x_0}(f(x) + g(x)) = L + M$
2. $\lim_{x\to x_0}(f(x)g(x)) = LM$

3. $\lim_{x \to x_0} \frac{f(x)}{g(x)} = \frac{L}{M}$, *provided* $M \neq 0$

4. $\lim_{x \to x_0} (cf(x)) = cL$ *for any constant* $c \in \mathbb{R}$

Definition 1.4 (Limit at infinity)**.** We say that $\lim_{x \to \infty} f(x) = \ell$ if for every $\varepsilon > 0$, there exists an $A > 0$ such that for all x,

$$x > A \implies |f(x) - \ell| < \varepsilon.$$

One makes the analogous definition is made for limits at $-\infty$.

Example 1.5. We prove $\lim_{x \to \infty} \frac{1}{x} = 0$. Given $\varepsilon > 0$, we require $\left|\frac{1}{x} - 0\right| < \varepsilon$, which means $\frac{1}{x} < \varepsilon$ for $x > 0$. Rearranging gives $x > \frac{1}{\varepsilon}$. We choose $A = \frac{1}{\varepsilon}$. For all $x > A$, we have $x > \frac{1}{\varepsilon}$, so $\frac{1}{x} < \varepsilon$.

Definition 1.6. We say that $\lim_{x \to x_0} f(x) = \infty$ if for every $B > 0$, there exists a $\delta > 0$ such that for all x,

$$0 < |x - x_0| < \delta \implies f(x) > B.$$

Again, one makes the analogous definition for when $\lim_{x \to x_0} f(x) = -\infty$, and when $x \to \pm\infty$.

Definition 1.7 (One-sided limits)**.** The right-sided limit $\lim_{x \to x_0^+} f(x) = \ell$ if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that for all x,

$$0 < x - x_0 < \delta \implies |f(x) - \ell| < \varepsilon.$$

The left-sided limit $\lim_{x \to x_0^-} f(x) = \ell$ if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that for all x,

$$0 < x_0 - x < \delta \implies |f(x) - \ell| < \varepsilon.$$

A two-sided limit exists if and only if both one-sided limits exist and are equal.

Example 1.8. Let $H : \mathbb{R} \to \mathbb{R}$ (the so-called Heaviside function) be defined by

$$H(x) = \begin{cases} 1, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

We see that $\lim_{x\to 0^-} H(x) = 0$ and $\lim_{x\to 0^+} H(x) = 1$, and $H(0) = 1$. Even though $H(0) = 1$, the limit approaching 0 from the left is still zero.

Theorem 1.9 (Squeeze Theorem)**.** *Suppose $g(x) \le f(x) \le h(x)$ for all x in an open interval containing x_0, except possibly at x_0 itself. If $\lim_{x\to x_0} g(x) = \lim_{x\to x_0} h(x) = \ell$, then $\lim_{x\to x_0} f(x) = \ell$.*

Example 1.10. Evaluate $\lim_{x\to 0} x\sin\left(\frac{1}{x}\right)$. Since $-1 \le \sin\left(\frac{1}{x}\right) \le 1$ for all $x \ne 0$, we multiply the inequality by $|x|$ to obtain:

$$-|x| \le x\sin\left(\frac{1}{x}\right) \le |x|.$$

We know $\lim_{x\to 0}(-|x|) = 0$ and $\lim_{x\to 0}|x| = 0$. By the comparison test, $\lim_{x\to 0} x\sin\left(\frac{1}{x}\right) = 0$.

Example 1.11. A famous limit that is an indeterminate form but can be derived using geometric ideas is the following:

$$\lim_{x\to 0}\frac{\sin x}{x}.$$

Consider a unit circle centered at the origin $O(0,0)$. Let x be a small positive angle such that $0 < x < \frac{\pi}{2}$, measured in radians. Let A be the point $(1,0)$ and B be the point on the circle such that $\angle AOB = x$. The coordinates of B are $(\cos x, \sin x)$.

Draw a vertical tangent line to the circle at point A. Extend the ray OB until it intersects this tangent line at point C. The coordinates of C are $(1, \tan x)$.

We compare the areas of three geometric regions:

1. The triangle $\triangle OAB$, which has base 1 and height $\sin x$.
2. The circular sector OAB, which has radius 1 and central angle x.
3. The triangle $\triangle OAC$, which has base 1 and height $\tan x$.

From the geometric construction, the triangle $\triangle OAB$ is contained within the sector OAB, which in turn is contained within the triangle $\triangle OAC$. Therefore, their areas satisfy the inequality:

$$\text{Area}(\triangle OAB) < \text{Area}(\text{sector } OAB) < \text{Area}(\triangle OAC).$$

Substituting the formulas for these areas yields:

$$\frac{1}{2}(1)(\sin x) < \frac{1}{2}(1)^2 x < \frac{1}{2}(1)(\tan x).$$

Multiplying by 2, we obtain:

$$\sin x < x < \tan x.$$

Since $0 < x < \frac{\pi}{2}$, we know $\sin x > 0$. Dividing the entire inequality by $\sin x$ gives:

$$1 < \frac{x}{\sin x} < \frac{1}{\cos x}.$$

Taking the reciprocal of all terms reverses the inequalities:

$$\cos x < \frac{\sin x}{x} < 1.$$

As $x \to 0^+$, we know $\lim_{x\to 0^+} \cos x = 1$ and $\lim_{x\to 0^+} 1 = 1$. By the Squeeze Theorem, it follows that:

$$\lim_{x\to 0^+} \frac{\sin x}{x} = 1.$$

Since $f(x) = \frac{\sin x}{x}$ is an even function $\left(\text{as } \frac{\sin(-x)}{-x} = \frac{-\sin x}{-x} = \frac{\sin x}{x}\right)$, the left-hand limit is identical to the right-hand limit:

$$\lim_{x\to 0^-} \frac{\sin x}{x} = 1.$$

Therefore, the two-sided limit evaluates to 1:

$$\lim_{x\to 0} \frac{\sin x}{x} = 1.$$

Example 1.12. Another simple trigonometric limit is

$$\lim_{x\to 0} \frac{\cos x - 1}{x},$$

which again is of an indeterminate form, and so care must be taken to find this limit.

We evaluate the limit by multiplying the numerator and denominator by the conjugate of the numerator, $\cos x + 1$:

$$\begin{aligned}\lim_{x\to 0} \frac{\cos x - 1}{x} &= \lim_{x\to 0} \frac{(\cos x - 1)(\cos x + 1)}{x(\cos x + 1)} \\ &= \lim_{x\to 0} \frac{\cos^2 x - 1}{x(\cos x + 1)}.\end{aligned}$$

Using the Pythagorean identity $\sin^2 x + \cos^2 x = 1$, we substitute $\cos^2 x - 1 = -\sin^2 x$:

$$\lim_{x\to 0} \frac{-\sin^2 x}{x(\cos x + 1)} = \lim_{x\to 0} \left(-\frac{\sin x}{x} \cdot \frac{\sin x}{\cos x + 1}\right).$$

Using the algebra of limits and the result from the previous proof:

$$\begin{aligned}\lim_{x\to 0}\left(-\frac{\sin x}{x}\cdot\frac{\sin x}{\cos x+1}\right) &= -\left(\lim_{x\to 0}\frac{\sin x}{x}\right)\cdot\left(\lim_{x\to 0}\frac{\sin x}{\cos x+1}\right)\\ &= -(1)\cdot\left(\frac{0}{1+1}\right)\\ &= 0.\end{aligned}$$

Therefore, $\lim_{x\to 0}\frac{\cos x-1}{x}=0$.

1.2 Continuity

Definition 1.13 (Continuity at a point)**.** A function f is continuous at x_0 if for every $\varepsilon>0$, there exists a $\delta>0$ such that for all x in the domain of f,

$$|x-x_0|<\delta \implies |f(x)-f(x_0)|<\varepsilon.$$

Equivalently, this requires that $\lim_{x\to x_0} f(x)=f(x_0)$.

Remark. For f to be continuous at x_0, three conditions must be satisfied: $f(x_0)$ must be defined, $\lim_{x\to x_0} f(x)$ must exist, and these two values must be equal.

Example 1.14. Let $f(x)$ be defined as:

$$f(x)=\begin{cases}\frac{x^2-1}{x-1} & x\neq 1\\ 2 & x=1\end{cases}$$

We check for continuity at $x=1$. We have $f(1)=2$. Taking the limit as $x\to 1$, we note that $x\neq 1$, so we can divide by $x-1$:

$$\lim_{x\to 1}\frac{x^2-1}{x-1}=\lim_{x\to 1}\frac{(x-1)(x+1)}{x-1}=\lim_{x\to 1}(x+1)=2.$$

Since $\lim_{x\to 1} f(x)=f(1)$, the function is continuous at $x=1$.

There is an analogous version of the Algebra of Limits for continuous functions (as one would expect, since continuity is defined in terms of limits):

Theorem 1.15 (Algebra of Continuous Functions)**.** *Let f,g be continuous functions. Then*

1. *$f+g$ is continuous.*

2. *fg is continuous.*

3. *$\frac{f}{g}$ is continuous, provided $g(x) \neq 0$ for all x.*

4. *$g \circ f$ is continuous.*

There are pathological examples of functions which are discontinuous everywhere on $\mathbb{R}$. For example, consider the function $f : \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

By density of the rational and irrational numbers in $\mathbb{R}$, one checks that f is discontinuous everywhere (exercise).

Chapter 2

The Derivative

Definition 2.1 (Differentiability)**.** A function f is differentiable at a point x if the following limit exists:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}.$$

For this limit to exist, both the left and right derivatives must exist and be equal.

There are alternative, equivalent ways to express the derivative limit. For instance, letting $y = x + h$, we can write:

$$f'(x) = \lim_{y \to x} \frac{f(y) - f(x)}{y - x}.$$

Additionally, if a function is already known to be differentiable at x, we can evaluate the derivative using the symmetric difference quotient:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x-h)}{2h}.$$

Remark. The existence of the symmetric limit does *not* imply differentiability. For example, if $f(x) = |x|$, the symmetric limit at $x = 0$ evaluates to 0, even though the standard derivative does not exist.

One can continue in this way and define *higher order derivatives*, $f^{(2)}, f^{(3)}, \ldots$ and so on, either indefinitely (in which case we say that the function is *infinitely differentiable* and is in C^∞) or up to some finite order, say, $n \in \mathbb{N}$.

Example 2.2. We define, for $n \geq 0$, the functions $f_n : \mathbb{R} \to \mathbb{R}$ given by

$$f_n(x) = \begin{cases} x^n \sin\left(\frac{1}{x}\right), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

We will analyse when f_n is differentiable, and for how many times. In the case $n = 0$, one checks that f_0 is not continuous at zero (observe that the function oscillates infinitely many times on any interval $(-\delta, \delta)$ about zero, so intuitively the function never settles on a number as $x \to 0$).

For $n = 1$, though continuous at zero, the function f_1 is not differentiable at 0 because when we evaluate the quotient, for $x \neq 0$, we have

$$\frac{f_1(x) - f_1(0)}{x - 0} = \frac{x \sin\left(\frac{1}{x}\right)}{x} = \sin\left(\frac{1}{x}\right),$$

which, as we just mentioned, does not have a limit as $x \to 0$.

For $n \geq 2$, the function is differentiable. Given $n \geq 2$, the quotient evalutes to

$$\frac{f_n(x) - f_n(0)}{x - 0} = x^{n-1} \sin\left(\frac{1}{x}\right),$$

and since $n - 1 > 0$, $x^{n-1} \to 0$ so using the fact that $|\sin x| \leq 1$ for all x, the squeeze theorem yields

$$\lim_{x \to 0} \frac{f_n(x) - f_n(0)}{x - 0} = 0$$

for $n \geq 2$.

As an exercise, one can in fact prove that f_n is differentiable exactly $n - 1$ times for $n \geq 2$.

Example 2.3. Let $f(x) = x^2$. We compute the derivative at x using the standard limit definition:

$$\begin{aligned} f'(x) &= \lim_{h \to 0} \frac{(x+h)^2 - x^2}{h} \\ &= \lim_{h \to 0} \frac{x^2 + 2xh + h^2 - x^2}{h} \\ &= \lim_{h \to 0} (2x + h) = 2x. \end{aligned}$$

Differentiability is stronger than continuity:

Theorem 2.4. *If $f(x)$ is differentiable at $x = x_0$, then it is also continuous there.*

Proof. Suppose f is differentiable at x_0. We want to show that $\lim_{x \to x_0} f(x) = f(x_0)$, which is equivalent to showing $\lim_{x \to x_0}(f(x) - f(x_0)) = 0$. We manipulate the limit:

$$\lim_{x \to x_0} (f(x) - f(x_0)) = \lim_{x \to x_0} \left(\frac{f(x) - f(x_0)}{x - x_0} \cdot (x - x_0) \right).$$

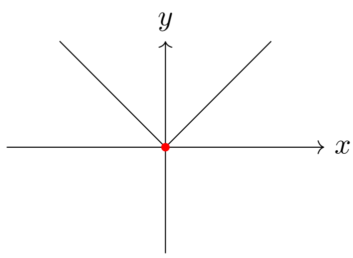
Using the algebraic properties of limits, this separates into:

$$\lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0} \cdot \lim_{x \to x_0} (x - x_0) = f'(x_0) \cdot 0 = 0.$$

Thus, $f(x) \to f(x_0)$ as $x \to x_0$, meaning f is continuous at x_0. □

Remark. The converse of this theorem is not true; a continuous function is not necessarily differentiable.

Example 2.5. Consider the function $f(x) = |x|$ at $x = 0$.



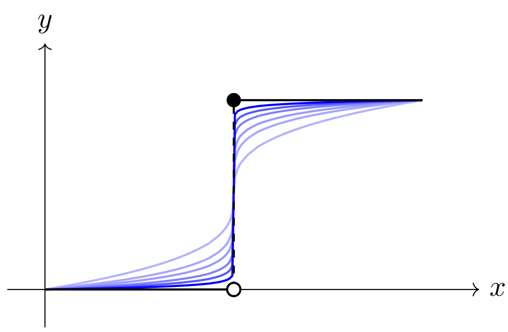
The function is continuous at $x = 0$ because $\lim_{x \to 0} |x| = 0 = f(0)$. However, evaluating the left and right derivatives yields:

$$\lim_{h \to 0^+} \frac{|0 + h| - |0|}{h} = \lim_{h \to 0^+} \frac{h}{h} = 1,$$

$$\lim_{h \to 0^-} \frac{|0 + h| - |0|}{h} = \lim_{h \to 0^-} \frac{-h}{h} = -1.$$

Since the left and right limits are not equal, the two-sided limit does not exist, and $f(x)$ is not differentiable at $x = 0$.

Consider functions with jump discontinuities. Attempting to approximate them with smooth, polynomial functions provides useful intuition for their derivatives. If we were to connect a discontinuous jump with an approximating continuous curve, the curve would require an incredibly steep slope downwards (or upwards) at the discontinuity. As the approximation becomes exact, the connecting slope becomes vertical, meaning the right (or left) slope does not exist and tends to $\pm\infty$. A slope of $\pm\infty$ represents a vertical tangent line. This idea is illustrated below.



Example 2.6. Suppose the position of a particle moving along a straight line at time t (in seconds) is given by the function $s(t) = t^2 + 3t$ (in meters).

The average speed of the particle over the time interval $[1, 1 + \Delta t]$ is the average rate of change of position:

$$v_{\text{avg}} = \frac{s(1+\Delta t) - s(1)}{\Delta t} = \frac{\big((1+\Delta t)^2 + 3(1+\Delta t)\big) - (1^2 + 3(1))}{\Delta t}.$$

Expanding the numerator yields:

$$v_{\text{avg}} = \frac{1 + 2\Delta t + (\Delta t)^2 + 3 + 3\Delta t - 4}{\Delta t} = \frac{5\Delta t + (\Delta t)^2}{\Delta t} = 5 + \Delta t \quad (\text{for } \Delta t \neq 0).$$

This tells us the average speed over any small interval starting at $t = 1$. To find the *instantaneous* speed of the particle at exactly $t = 1$, we take the limit as the interval length Δt approaches 0:

$$v(1) = s'(1) = \lim_{\Delta t \to 0} (5 + \Delta t) = 5 \text{ m/s}.$$

Example 2.7. Geometrically, the derivative represents the slope of the tangent line to a curve at a given point. Consider the curve defined by $f(x) = x^3 - 2x$. We wish to find the equation of the tangent line to this curve at $x = 2$.

First, we find the coordinates of the point of tangency by evaluating the function:

$$f(2) = (2)^3 - 2(2) = 8 - 4 = 4.$$

So, the line passes through the point $(2, 4)$. Next, we calculate the derivative $f'(x)$ to find the formula for the slope at any point x:

$$f'(x) = \frac{\mathrm{d}}{\mathrm{d}x}(x^3 - 2x) = 3x^2 - 2.$$

Evaluating the derivative at $x = 2$ gives the slope m of the tangent line:

$$m = f'(2) = 3(2)^2 - 2 = 12 - 2 = 10.$$

Using the point-slope form of a linear equation, $y - y_1 = m(x - x_1)$, the equation of the tangent line is:

$$y - 4 = 10(x - 2) \implies y = 10x - 16.$$

2.1 Properties of the Derivative

Proposition 2.8. *Let f, g be differentiable functions. Then*

1. *(Linearity): $f + g$ is differentiable with $(f + g)' = f' + g'$.*

2. *(Linearity): cf is differentiable for all $c \in \mathbb{R}$ with $(cf)' = cf'$.*
3. *(Product Rule): fg is differentiable with $(fg)' = f'g + fg'$.*
4. *(Quotient Rule): $\frac{f}{g}$ is differentiable, provided $g(x) \neq 0$ for all x, with $\left(\frac{f}{g}\right)' = \frac{gf' - fg'}{g^2}$.*

Example 2.9. As a consequence of the above results, polynomials are differentiable, with

$$\frac{\mathrm{d}}{\mathrm{d}x} \sum_{k=0}^{n} a_k x^k = \sum_{k=1}^{n} k a_k x^{k-1}.$$

Theorem 2.10 (Chain Rule)**.** *Let f, g be differentiable such that g is defined on the range of f. Then $g \circ f$ is differentiable with*

$$\frac{\mathrm{d}}{\mathrm{d}x} g(f(x)) = g'(f(x)) f'(x)$$

for all x.

Example 2.11. Let $f(x) = \sin\left(\sqrt{1+x^2}\right)$. Using the chain rule twice (first on $\sin(g(x))$ where $g(x) = \sqrt{1+x^2}$, and then on $g(x) = \sqrt{h(x)}$ where $h(x) = 1 + x^2$), we get

$$f'(x) = \frac{\mathrm{d}}{\mathrm{d}x} \sin\left(\sqrt{1+x^2}\right) = \cos\left(\sqrt{1+x^2}\right) (\sqrt{1+x^2})' = \frac{x \cos\left(\sqrt{1+x^2}\right)}{\sqrt{1+x^2}}.$$

2.2 Implicit Differentiation

A useful consequence of the chain rule is the concept of implicit differentiation.

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Chapter 3

IVT and MVT

In this chapter we will introduce some important theorems regarding the theory of continuous and differentiable real-valued functions. Many of these are intuitive and should be familiar to the reader from learning calculus previously.

3.1 Extrema and Critical Points

One important problem in calculus is that of finding critical points and classifying (local and global) extrema that certain functions possess. It is a common scenario in applied mathematics to be faced with a function (for example, a profit function in economics) whose maximum or minimum value on its domain (i.e. global extrema) we seek the values of. There are some subtleties regarding the existence and classification of such points, and we aim to clarify such matters in this section.

Definition 3.1 (Global and Local Extrema)**.** Let f be defined on a domain D.

1. f has an absolute (or global) maximum at $c \in D$ if $f(c) \geq f(x)$ for all $x \in D$.
2. f has a local maximum at c if there exists an open interval I containing c such that $f(c) \geq f(x)$ for all $x \in I \cap D$.

Minimums are defined analogously.

Example 3.2. The function $f(x) = \sin x$ on the domain $\mathbb{R}$ has infinitely many global minima and global maxima, occuring at values of x where $f(x) = -1$ and $f(x) = 1$ respectively. These values of x are precisely

$$x = \frac{3\pi}{2} + 2n\pi \quad \text{and} \quad x = \frac{\pi}{2} + 2n\pi, \quad n \in \mathbb{Z},$$

respectively.

One makes the natural definitions for *strict* global extrema, which are unique, if they exist.

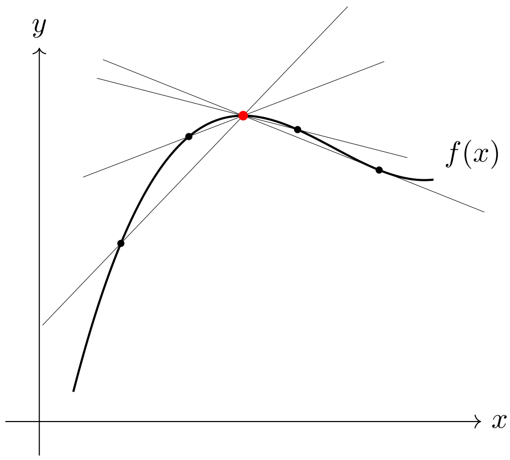
Definition 3.3 (Critical Point)**.** A point c in the domain of a differentiable function f is called a critical point if either $f'(c) = 0$.

Critical points of a differentiable function are points where the tangent line to the graph of f is horizontal. However, this does *not* mean that a local extremum exists there; for example, consider the function $f(x) = x^3$ on $\mathbb{R}$. At $x = 0$, $f'(0) = 0$, however the function f is strictly increasing on $\mathbb{R}$, so f admits no local or global extrema.

However, the converse is true: if a function has a local extremum at a point, and it is differentiable there, then the derivative vanishes at that point. This is called Fermat's Theorem:

Theorem 3.4 (Fermat's Theorem)**.** *Let f be a function which is defined and differentiable on the open interval (a, b). If $c \in (a, b)$ is a maximum (or minimum) for the function, then $f'(c) = 0$.*

The idea is that at, say, a local maximum, the slopes from the left are positive and the slopes from the right are negative (see the picture below), and if f is differentiable there, then these slopes are both converging to the same value, which must be zero by the squeeze theorem.



Example 3.5. Let $f(x) = x^3 - 3x$. We find the critical points by solving $f'(x) = 0$. We have $f'(x) = 3x^2 - 3 = 3(x - 1)(x + 1)$. Thus, $x = 1$ and $x = -1$ are critical points. Evaluating the function, $f(1) = -2$ (a local minimum) and $f(-1) = 2$ (a local maximum).

3.2 The Mean Value Theorem

Two results that are used to prove the Mean Value Theorem (MVT) are the Extreme Value Theorem and Rolle's Theorem (both below). Again, we prove these rigorously in Analysis I, so they will not be proved again here.

Theorem 3.6 (Extreme Value Theorem). *If a function f is continuous on a closed and bounded interval $[a, b]$, then f attains an absolute maximum value $f(c)$ and an absolute minimum value $f(d)$ at some numbers $c, d \in [a, b]$.*

Theorem 3.7 (Rolle's Theorem). *Let f be continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b). If $f(a) = f(b)$, then there exists $c \in (a, b)$ such that $f'(c) = 0$.*

Theorem 3.8 (Mean Value Theorem). *Let f be continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b). Then there exists $c \in (a, b)$ such that*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Example 3.9. Consider $f(x) = x^2$ on the interval $[1, 3]$. The function is continuous on $[1, 3]$ and differentiable on $(1, 3)$. The average rate of change is $\frac{f(3)-f(1)}{3-1} = \frac{9-1}{2} = 4$. By the Mean Value Theorem, there exists $c \in (1, 3)$ such that $f'(c) = 4$. Since $f'(x) = 2x$, solving $2c = 4$ yields $c = 2$, which lies in $(1, 3)$.

Definition 3.10. Let $f : I \to \mathbb{R}$ be a function. We say that f is *increasing* if for all $x_1 < x_2$ in I we have $f(x_1) \leq f(x_2)$, and *decreasing* if for all $x_1 < x_2$ in I we have $f(x_1) \geq f(x_2)$.

One makes the natural definitions for *strictly* increasing and decreasing functions.

A nice application of the Mean Value Theorem is the following theorem regarding the relationship between the sign of the derivative and monotonicity of differentiable functions.

Theorem 3.11 (Monotonicity and Derivatives)**.** *Let f be differentiable on an interval I.*

1. *If $f'(x) > 0$ for all $x \in I$, then f is strictly increasing on I.*
2. *If $f'(x) < 0$ for all $x \in I$, then f is strictly decreasing on I.*
3. *If $f'(x) = 0$ for all $x \in I$, then f is constant on I.*

Proof. Let $x < y$ in I be arbitrary. Since f is differentiable on I, in particular f is continuous on $[x, y]$ and differentiable on (x, y), so by the Mean Value Theorem there exists $c \in (x, y)$ such that

$$f'(c) = \frac{f(y) - f(x)}{y - x}.$$

Since $y - x > 0$, we observe that if $f'(t) > 0$ for all $t \in I$, then in particular $f'(c) > 0$ so $f(y) > f(x)$, so f is strictly increasing on I since x and y were arbitrary. Similarly if $f'(t) < 0$ for all $t \in I$ then f is strictly decreasing on I. If $f'(t) = 0$ for all $t \in I$ then $f(x) = f(y)$ for all x and y in I, i.e. f is constant on I. $\square$

Example 3.12. We will prove the inequality $\sin x \leq x$ for $x \geq 0$. Define $g : [0, \infty) \to \mathbb{R}$ by $g(x) = x - \sin x$; then $g(0) = 0$ and g is differentiable with

$$g'(x) = 1 - \cos x$$

for all $x \geq 0$. Since $|\cos x| \leq 1$ for all $x \in \mathbb{R}$, we have $g'(x) \geq 0$ for all $x \geq 0$, so the function g is increasing on $[0, \infty)$, so since $g(0) = 0$, $g(x) \geq 0$ for all $x \geq 0$, i.e. $\sin x \leq x$ for all $x \geq 0$, as required.

3.3 Intermediate Value Theorem

A very intuitive theorem, but one that is not so easy to prove (again, we prove this theorem rigorously in Analysis I) is the Intermediate Value Theorem (IVT), which intuitively states that continuous functions cannot "jump" over values.

Theorem 3.13 (Intermediate Value Theorem)**.** *Let f be a continuous function on the closed interval $[a, b]$. If y is any number strictly between $f(a)$ and $f(b)$, then there exists $c \in (a, b)$ such that $f(c) = y$.*

Practically, one example where this theorem is useful is for estimating when roots of continuous functions exist.

Example 3.14. We show that the equation $x^3 - x - 1 = 0$ has a root in the interval $[1, 2]$. Let $f(x) = x^3 - x - 1$. Since f is a polynomial, it is continuous everywhere. We evaluate the endpoints: $f(1) = 1 - 1 - 1 = -1$ and $f(2) = 8 - 2 - 1 = 5$. Since $f(1) < 0 < f(2)$, the Intermediate Value Theorem guarantees the existence of a $c \in (1, 2)$ such that $f(c) = 0$.